CONDITIONS	RULES				
OUTCOMES	Yes. The patient has evidence of immunity.	No. The patient does not have evidence of immunity.	Yes. The patient has evidence of immunity.	No. The patient does not have evidence of immunity.	No. The patient does not have evidence of immunity.

7.3 Determine Contraindications

Determine contraindications assesses if any or all series for an antigen are contraindicated for the patient. Contraindications may be applied at either the antigen or vaccine level.

Given the complex nature of contraindications, it may not always be possible to conclusively determine if a contraindication applies to a patient. To minimize missed doses, in the case where a contraindication cannot be definitively determined to be relevant for a patient, the contraindication will not be applied, but a notification should be made to a clinician alerting them to the presence of the possible contraindication which could not be resolved.

The following process model is used to assess contraindications.

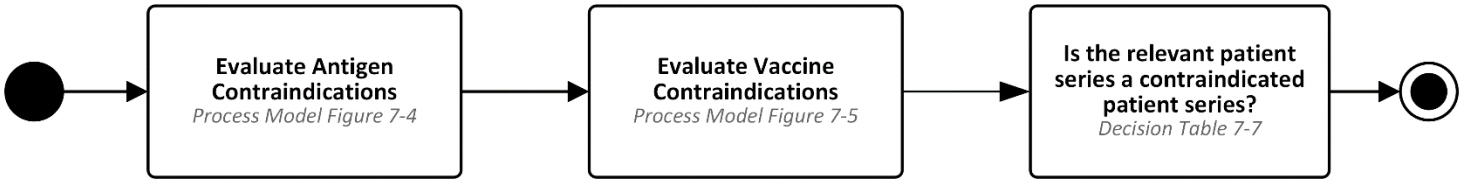

FIGURE 7-3 CONTRAINDICATION PROCESS MODEL

TABLE 7-4 DETERMINE CONTRAINDICATION ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Patient	Active Patient Observations	-
Patient	Adverse Reactions	-
Supporting Data	Contraindication elements	-
Processing data	Assessment Date	current date
Calculated date (CALCDTCI-1)	Contraindication Begin Age Date	01/01/1900
Calculated date (CALCDTCI-2)	Contraindication End Age Date	12/31/2999

An antigen contraindication prevents all relevant patient series for that antigen from recommending further vaccination for the patient. That is, no relevant patient series for the antigen should forecast dates for the patient. The patient series status will be contraindicated for each relevant patient series.

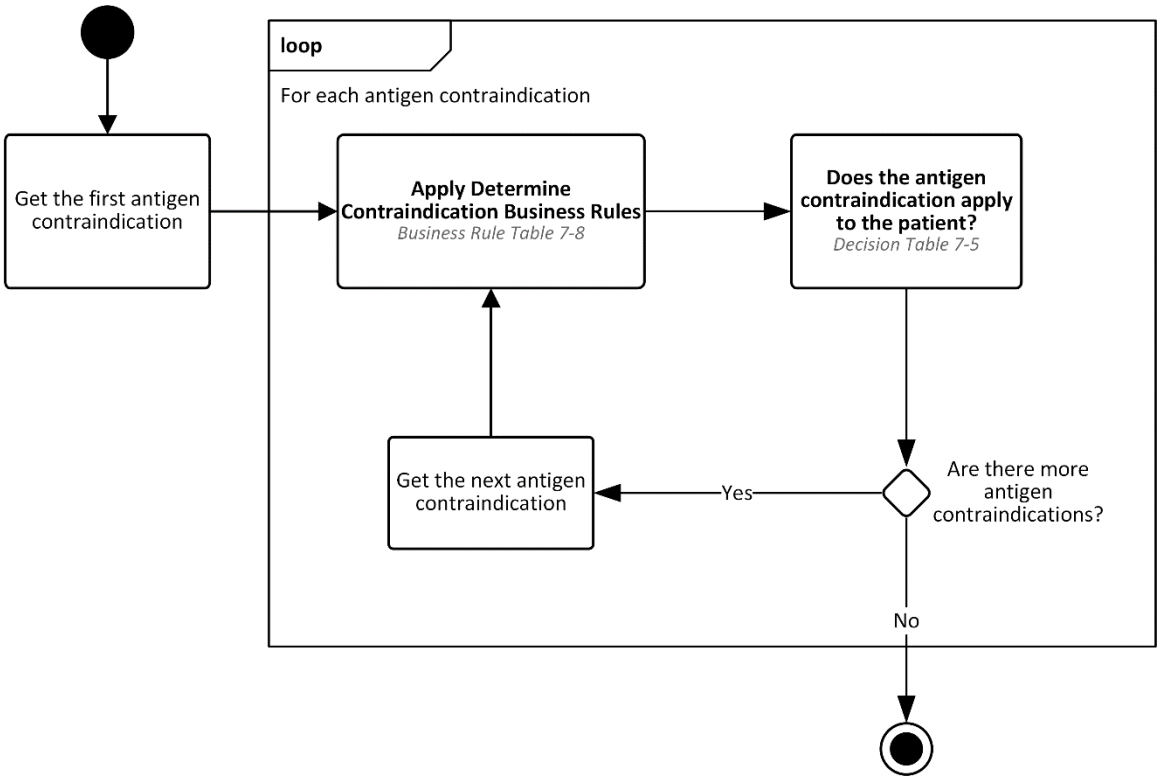

FIGURE 7-4 ANTIGEN CONTRAINDICATION PROCESS MODEL

TABLE 7-5 DOES THE ANTIGEN CONTRAINDICATION APPLY TO THE PATIENT?

CONDITIONS	RULES				
Does the antigen contraindication describe any active patient observations?	Yes	No	No	Unknown	-
Does the antigen contraindication describe any adverse reactions?	No	Yes	No	Unknown	-
Is the contraindication begin age date ≤ assessment date < contraindication end age date?	Yes	Yes	Yes	Yes	No

CONDITIONS	RULES				
OUTCOMES	Yes. The antigen contraindication applies to the patient.	Yes. The antigen contraindication applies to the patient.	No. The antigen contraindication does not apply to the patient.	No. It could not be determined if the antigen contraindication applies to the patient; however, the Contraindication Text Description should be made available to the clinician for manual determination.	No. The antigen contraindication does not apply to the patient.

A vaccine contraindication eliminates a specific vaccine from being forecast for the patient.

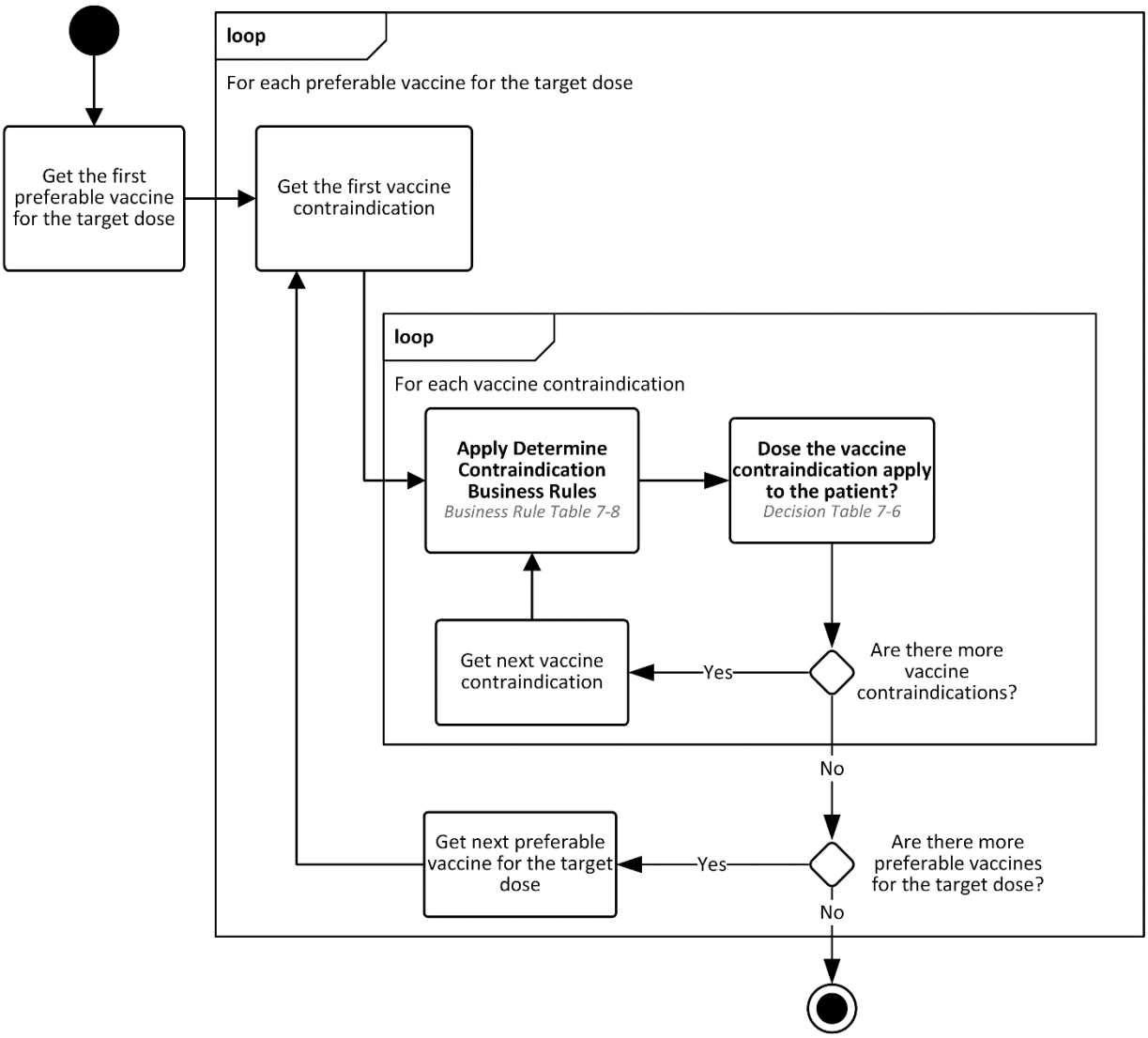

FIGURE 7-5 VACCINE CONTRAINDICATION PROCESS MODEL

TABLE 7-6 DOES THE VACCINE CONTRAINDICATION APPLY TO THE PATIENT?

CONDITIONS	RULES						
Does the vaccine contraindication describe any active patient observations?	Yes	Yes	No	No	No	-	Unknown
Does the vaccine contraindication describe any adverse reactions?	No	No	Yes	Yes	No	-	Unknown
Is the contraindication begin age date ≤ assessment date < contraindication end age date?	Yes	Yes	Yes	Yes	-	No	Yes
Is the vaccine type of the preferable vaccine one of the contraindicated vaccine types for the contraindication?	Yes	No	Yes	No	-	-	Yes
OUTCOMES	Yes. The vaccine contraindication applies to the patient.	No. The vaccine contraindication does not apply to the patient.	Yes. The vaccine contraindication applies to the patient.	No. The vaccine contraindication does not apply to the patient.	No. The vaccine contraindication does not apply to the patient.	No. The vaccine contraindication does not apply to the patient.	No. It could not be determined if the vaccine contraindication applies to the patient; however, the Contraindication Text Description should be made available to the clinician for manual determination.

A relevant patient series should not be forecast if either an antigen contraindication exists, or if all preferable vaccines are contraindicated.

TABLE 7-7 IS THE RELEVANT PATIENT SERIES A CONTRAINDICATED PATIENT SERIES?

CONDITIONS	RULES		
Are there any antigen contraindication that apply to the patient?	Yes	No	No

CONDITIONS	RULES		
Do all preferable vaccines for the relevant patient series have at least one vaccine contraindication that applies to the patient?	-	Yes	No
OUTCOMES	Yes. The relevant patient series is a contraindicated patient series.	Yes. The relevant patient series is a contraindicated patient series.	No. The relevant patient series is not a contraindicated patient series.

TABLE 7-8 DETERMINE CONTRAINDICATION BUSINESS RULES

Business Rule ID	Business Rule
CALCDTCI-1	A patient's contraindication begin age date must be calculated as the patient's date of birth plus the contraindication begin age of a contraindication.
CALCDTCI-2	A patient's contraindication end age date must be calculated as the patient's date of birth plus the contraindication end age of a contraindication.

7.4 Determine Forecast Need

Determine forecast need determines if there is a need to forecast dates. This involves reviewing patient data, antigen administered records, and patient series. This is a prerequisite before a CDS engine can produce forecast dates and reasons.

The following process model, attribute table, and decision table are used to determine the need to generate forecast dates.

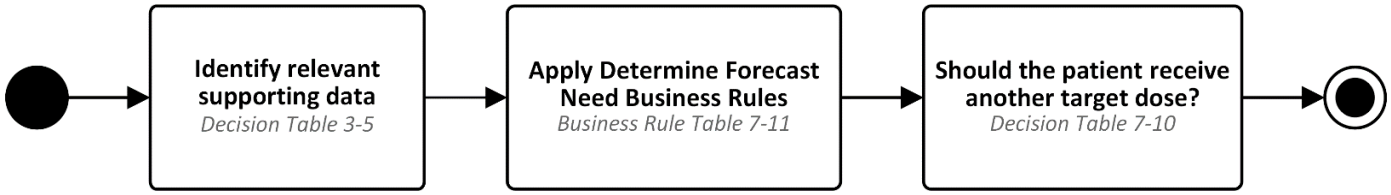

FIGURE 7-6 DETERMINE FORECAST NEED PROCESS MODEL

TABLE 7-9 DETERMINE FORECAST NEED ATTRIBUTES

Attribute Type	Attribute Name	Assumed Value if Empty
Immunization history	Vaccine Dose(s) Administered	-
Relevant Patient series	Target Dose Statuses	-
Supporting Data (Seasonal Recommendation)	Seasonal Recommendation End Date	12/31/2999
Section 7.2 Outcome	Evidence of Immunity	No evidence of immunity
Section 7.3 Outcome	Contraindicated Patient Series	The relevant patient series is NOT a contraindicated patient series.